APPENDIX
Math Compendium

A Notation and Nomenclature

$x(t)$, $y(t)$: continuous-time signals	$\in$: element of (belongs to)
$x[n]$, $y[n]$: discrete-time signals	$\mathbb{N}$: set of natural numbers: 0, 1, 2, 3, …
$X(\omega)$, $X(z)$, $X(s)$: transformed signals	$\mathbb{Z}$: set of integer numbers: 0, ±1, ±2, …
N, T: signal period (N_0, T_0: fundamental period)	$\mathbb{Q}$: set of rational numbers (fractions)
f: signal frequency (f_0: fundamental frequency)	$\mathbb{R}$: set of real numbers ($\mathbb{R}^+$ positive)
ω: angular frequency (ω_0: specific angular frequency)	$\mathbb{C}$: set of complex numbers
H, G: transfer function; system operator	$\forall$: for all (any)
T_s: sampling interval	$\exists$: exists
θ: signal phase	$\equiv$: identically equal
$\mathrm{sinc}(t) = \dfrac{\sin(t)}{t}$	$\mathrm{sinc}_\pi(t) = \dfrac{\sin(\pi t)}{\pi t}$
$\mathcal{U}(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$\mathrm{ramp}(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$
$\mathrm{rect}(t) = \mathcal{U}(t+0.5) - \mathcal{U}(t-0.5) = \begin{cases} 1, & \lvert t\rvert \leq 0.5 \\ 0, & \lvert t\rvert > 0.5 \end{cases}$	$\mathrm{tri}(t) = \begin{cases} 1-\lvert t\rvert, & \lvert t\rvert < 1 \\ 0, & \lvert t\rvert \geq 1 \end{cases}$

B Trigonometric Identities

1	$\cos(\theta_1 \pm \theta_2) = \cos(\theta_1)\cos(\theta_2) \mp \sin(\theta_1)\sin(\theta_2)$
2	$\sin(\theta_1 \pm \theta_2) = \sin(\theta_1)\cos(\theta_2) \pm \cos(\theta_1)\sin(\theta_2)$
3	$2\cos(\theta_1)\cos(\theta_2) = \cos(\theta_1 - \theta_2) + \cos(\theta_1 + \theta_2)$
4	$2\sin(\theta_1)\sin(\theta_2) = \cos(\theta_1 - \theta_2) - \cos(\theta_1 + \theta_2)$
5	$2\sin(\theta_1)\cos(\theta_2) = \sin(\theta_1 - \theta_2) + \sin(\theta_1 + \theta_2)$
6	$\cos\left(\theta \pm \dfrac{\pi}{2}\right) = \mp\sin(\theta)$
7	$\sin\left(\theta \pm \dfrac{\pi}{2}\right) = \pm\cos(\theta)$

8	$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$
9	$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$
10	$2\cos^2(\theta) = 1 + \cos(2\theta)$
11	$2\sin^2(\theta) = 1 - \cos(2\theta)$
12	$4\cos^3(\theta) = 3\cos(\theta) + \cos(3\theta)$
13	$4\sin^3(\theta) = 3\sin(\theta) - \sin(3\theta)$
14	$\cos^2(\theta) + \sin^2(\theta) = 1$

C Complex Numbers

$j = \sqrt{-1}; \quad j^2 = -1; \quad j^3 = -j; \quad j^4 = 1; \quad j^5 = j; \ldots$

Cartesian: $s = a + bj$	**Polar:** $s = Re^{j\theta}$
Real value: $Re\{s\} = a$	Real value: $Re\{s\} = R\cos\theta$
Imaginary value: $Im\{s\} = b$	Imaginary value: $Im\{s\} = R\sin\theta$
Absolute value (magnitude): $\lvert s\rvert = \sqrt{a^2 + b^2}$	Absolute value (magnitude): $\lvert s\rvert = R$
Angle: $\angle s = \tan^{-1}\left(\frac{b}{a}\right)$	Angle: $\angle s = \theta$
Conjugate: $\overline{s} = a - bj$	Conjugate: $\overline{s} = Re^{-j\theta}$

C.1 Euler Theorem and de Moivre's Formula

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$	$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$	$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$
$\lvert e^{\pm j\theta}\rvert = 1$	$\angle e^{\pm j\theta} = \pm\theta$	$(re^{j\theta})^{\frac{1}{n}} = r^{1/n}e^{\frac{j(\theta+2\pi k)}{n}}; \quad n = \mathbb{N}, k = 0, \ldots, n-1$
$e^{\pm j\pi/2} = \pm j$	$e^{\pm j2n\pi} = 1; \quad n \in \mathbb{N}$	$e^{\pm j(2n+1)\pi} = -1; \quad n \in \mathbb{N}$

C.2 Multiplication

$$(a + bj)(c + dj) = (ac - bd) + (ad + bc)j$$

$$R_1e^{j\theta_1}R_2e^{j\theta_2} = R_1R_2e^{j(\theta_1+\theta_2)}$$

C.3 Division

$$\frac{a + bj}{c + dj} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}j$$

$$R_1e^{j\theta_1}/R_2e^{j\theta_2} = \frac{R_1}{R_2}e^{j(\theta_1-\theta_2)}$$

D Algebra

D.1 Inequalities

- Triangle inequality: $\forall a, b \in \mathbb{R}$: $\lvert a + b\rvert \le \lvert a\rvert + \lvert b\rvert$
  This inequality could be generalized to any finite number of a_i:

$$\left|\sum_{i=1}^{n} a_i\right| \le \sum_{i=1}^{n} |a_i|$$

  The equality reached only when a_i are all positive or all negative.

- $\forall a, b \in \mathbb{R}$: $|a-b| \geq ||a|-|b||$.
- $\forall a, b \in \mathbb{R}$: $a^2 + b^2 \geq 2|ab|$ (in particular from $(a-b)^2 \geq 0$ follows $a^2 + b^2 \geq 2ab$). The equality is reached if and only if $|a| = |b|$.
- $\forall a, b \in \mathbb{R}$ of the same sign ($ab > 0$):

$$\frac{a}{b} + \frac{b}{a} \geq 2$$

  and the equality is reached if and only if $a = b$.
- Cauchy inequality for finite number of $a_i \geq 0$:

$$\frac{a_1 + a_2 + \cdots + a_n}{n} = \frac{1}{n}\sum_{i=1}^{n} a_i \geq \sqrt[n]{\prod_{i=1}^{n} a_i} = \sqrt{a_1 a_2 \cdots a_n}$$

  which means that the geometric mean is not greater than the arithmetic mean.
- Hölder's inequality: $\forall a_i, b_i \in \mathbb{R}$, $\forall p > 1$:

$$\left|\sum_{i=1}^{n} a_i b_i\right| \leq \left(\sum_{i=1}^{n} |a_i|^p\right)^{\frac{1}{p}} \left(\sum_{i=1}^{n} |a_i|^{\frac{p}{p-1}}\right)^{\frac{p-1}{p}}$$

D.2 Polynomials

Polynomial of the free variable s is the sum of powers of s multiplied by constant coefficients:

$$P(s) = \sum_{k=0}^{n} a_k s^k = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

where a_k ($k = 0, 1, \ldots, n$) are constant coefficients and n is the order of the polynomial.

Theorem

Any polynomial of the order n has exactly n complex roots and it could be factorized as follows:

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = a_n (s - s_1)(s - s_2) \cdots (s - s_n),$$

where $s_1, \ldots, s_n$ are the roots.

Example

$$2s^2 + 6s + 4 = 2(s+2)(s+1)$$

where -1 and -2 are the roots of the polynomial $2s^2 + 6s + 4$.

Vieta's Formulas

For a polynomial $P(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$,

$$\begin{cases} s_1 + s_2 + \cdots + s_n = -\dfrac{a_{n-1}}{a_n} \\ (s_1 s_2 + s_1 s_3 + s_1 s_n) + (s_2 s_3 + s_2 s_4 + \cdots + s_2 s_n) + \cdots + s_{n-1} s_n = \dfrac{a_{n-2}}{a_n} \\ \vdots \\ s_1 s_2 \ldots s_n = (-1)^n \dfrac{a_0}{a_n} \end{cases}$$

In other words, the sum and product of roots could be derived from the coefficients without computing the roots.

Example

The sum of roots of $2s^2 + 6s + 4 = 0$ is $-\dfrac{6}{2} = -3$ and the product (multiplication) is $\dfrac{4}{2} = 2$.

E Calculus

E.1 Derivative Rules

Chain rule	$\dot{f}(g(t)) = \dfrac{d}{dt} f(g(t)) = \dfrac{df(g)}{dg}\dot{g}(t)$
Linearity	$\dfrac{d}{dt}(af(t) + bg(t)) = a\dot{f}(t) + b\dot{g}(t)$
Product rule	$\dfrac{d}{dt}(f(t)g(t)) = \dot{f}(t)g(t) + f(t)\dot{g}(t)$
Quotient rule	$\dfrac{d}{dt}\left(\dfrac{f(t)}{g(t)}\right) = \dfrac{\dot{f}(t)g(t) - f(t)\dot{g}(t)}{g^2(t)}$
Leibniz's rule	$\dfrac{d}{dt}\left[\int_{a(t)}^{b(t)} f(\lambda, t)\,d\lambda\right] = f(b(t), t)\dfrac{db(t)}{dt} - f(a(t), t)\dfrac{da(t)}{dt} + \int_{a(t)}^{b(t)} \dfrac{\partial f(\lambda, t)}{\partial t}\,d\lambda$

E.2 Derivatives Table

1	$\dfrac{d}{dt}(t^n) = nt^{n-1}$
2	$\dfrac{d}{dt}(e^{at}) = ae^{at}$
3	$\dfrac{d}{dt}(\ln t) = \dfrac{1}{t}$

4	$\dfrac{d}{dt}(\sin at) = a\cos at$
5	$\dfrac{d}{dt}(\cos at) = -a\sin at$
6	$\dfrac{d}{dt}(a^t) = a^t \ln a$

E.3 Integration

Linearity	$\int af(t)+bg(t)dt = a\int f(t)dt + b\int g(t)dt$
Integration by parts	$\int_a^b f(t)\dot{g}(t)\,dt = f(t)g(t)\Big\|_a^b - \int_a^b \dot{f}(t)g(t)\,dt$
Change in variable	$\lambda=\varphi(t)$: $\int_a^b f(\lambda)\,d\lambda = \int_{\varphi^{-1}(a)}^{\varphi^{-1}(b)} f(\varphi(t))\dot{\varphi}(t)\,dt$

E.4 Indefinite Integrals

1	$\int t^n dt = \frac{t^{n+1}}{n+1};\quad n\neq -1$
2	$\int \frac{1}{t}dt = \ln\lvert t\rvert$
3	$\int a^t dt = \frac{a^t}{\ln a}$
4	$\int e^{at}\,dt = \frac{e^{at}}{a}$
5	$\int te^{at}dt = e^{at}\left(\frac{t}{a}-\frac{1}{a^2}\right)$
6	$\int \cos(at)\,dt = \frac{1}{a}\sin(at)$
7	$\int \sin(at)\,dt = -\frac{1}{a}\cos(at)$
8	$\int t\cos(at)\,dt = \frac{1}{a^2}(\cos(at)+at\,\sin(at))$
9	$\int t\sin(at)\,dt = \frac{1}{a^2}(\sin(at)-at\cos(at))$

10	$\int t^n \cos t\,dt = x^n \sin t - n\int t^{n-1}\sin t\,dt$
11	$\int t^n \sin t\,dt = -t^n\cos t + n\int t^{n-1}\cos t\,dt$
12	$\int t^n e^{at}dt = \frac{1}{a}\left[t^n e^{at} - n\int t^{n-1}e^{ax}dt\right]$
13	$\int e^{at}\sin bt\,dt = \frac{e^{at}}{a^2+b^2}(a\sin bt - b\cos bt)$
14	$\int e^{at}\cos bt\,dt = \frac{e^{at}}{a^2+b^2}(a\cos bt + b\sin bt)$
15	$\int \cos(at)\cos(bt)\,dt = \frac{\sin[(a-b)t]}{2(a-b)} + \frac{\sin[(a+b)t]}{2(a+b)}$
16	$\int \sin(at)\sin(bt)\,dt = \frac{\sin[(a-b)t]}{2(a-b)} - \frac{\sin[(a+b)t]}{2(a+b)}$
17	$\int \sin(at)\cos(bt)dt = -\frac{\cos[(a-b)t]}{2(a-b)} - \frac{\cos[(a+b)t]}{2(a+b)}$
18	$\int \cos^2(at)\,dt = \frac{1}{4a}[2at+\sin(2at)]$

E.5 Definite Integrals

1	$\int_0^\infty \frac{t^{m-1}}{t^n+1}dt = \frac{\pi/n}{\sin(m\pi/n)}$
2	$\int_0^\infty t^n e^{-at}\,dt = \frac{n!}{a^{n+1}};\quad a>0$
3	$\int_0^\infty e^{-at}\cos bt\,dt = \frac{a}{a^2+b^2};\quad a>0$
4	$\int_0^\infty e^{-at}\sin bt\,dt = \frac{b}{a^2+b^2};\quad a>0$

5	$\int_0^\infty e^{-a^2t^2}\cos bt\,dt = \frac{\sqrt{\pi}}{2a}e^{-b^2/(4a^2)};\quad a>0$
6	$\int_0^\infty \frac{\sin at}{t}dt = \frac{\pi}{2}\,\mathrm{sign}(a)$
7	$\int_0^\infty \left(\frac{\sin t}{t}\right)^2 dt = \frac{\pi}{2}$
8	$\int_{-\infty}^\infty e^{\pm j2\pi at}\,dt = \delta(a)$

E.6 Finite Series

1	$\sum_{n=1}^{N} n = \frac{N(N+1)}{2}$
2	$\sum_{n=1}^{N} n^2 = \frac{N(N+1)(2N+1)}{6}$
3	$\sum_{n=1}^{N} n^3 = \frac{N^2(N+1)^2}{4}$

4	$\sum_{n=0}^{N} a^n = \frac{a^{N+1}-1}{a-1}; \quad a \neq 1$
5	$\sum_{n=0}^{N} \frac{N!}{(N-n)!n!} a^n b^{N-n} = (a+b)^N$
6	$\sum_{n=0}^{N} na^n = \frac{1-(N+1)a^N + Na^{N+1}}{(1-a)^2}$

E.7 Infinite Series

Taylor Series

$$f(t) = \sum_{n=0}^{\infty} \left(\frac{1}{n!} \frac{d^n}{dt^n} f(a) \right) (t-a)^n = f(a) + \dot{f}(a)(t-a) + \frac{1}{2}\ddot{f}(a)(t-a)^2 + \frac{1}{6}\dddot{f}(a)(t-a)^3 + \cdots$$

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!} = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} + \cdots$$

$$te^t = \sum_{n=0}^{\infty} \frac{t^n}{(n-1)!} = t + t^2 + \frac{t^3}{2} + \frac{t^4}{6} + \cdots$$

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n = 1 + t + t^2 + t^3 + t^4 + \cdots; \quad |t| < 1$$

$$\ln(1+t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{t^n}{n} = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \cdots$$

$$\cos t = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!} = 1 - \frac{t^2}{2} + \frac{t^4}{24} - \frac{t^6}{720} + \cdots$$

$$\sin t = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!} = t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040} + \cdots$$

Multivariable Taylor Series

A multivariable function $f(t)$ around the point a is approximated by

$$f(t+a) = f(a) + (\nabla f(a))^T t + \frac{1}{2} t^T \nabla^2 f(a) t + (\text{higher-order terms})$$

where both a and t are vectors, ∇ denotes the gradient (vector of partial derivatives of f), and ∇^2 denotes the Hessian matrix of second-order derivatives.

F Signals and Systems

F.1 Linear Time Invariant Systems

Let's assume that the response of a system to input $u_1(t)$ is $y_1(t)$ and the response to another input $u_2(t)$ is $y_2(t)$. The *system is linear* if for any u_1 and u_2 and for any scalars α and β, the response of this system to $\alpha u_1(t)+\beta u_2(t)$ is $\alpha y_1(t)+\beta y_2(t)$. In other words, linearity is when you multiply the input by some constant, the output is multiplied by the same constant, and the response to sum of signals is the sum of responses.

A system is *time invariant* if for any scalar τ the response of this system to $u_1(t-\tau)$ is $y_1(t-\tau)$. In other words, time invariance means that the response to a shifted signal is shifted response (by the same time).

F.2 Fourier Series

Trigonometric Fourier series for a function $x(t)$ defined on interval $[a,\ a+T]$:

$$x(t)=\frac{a_0}{2}+\sum_{k=1}^{\infty}\left[a_k\cos\left(k\frac{2\pi}{T}t\right)+b_k\sin\left(k\frac{2\pi}{T}t\right)\right]$$

where $a_k,\ b_k\in\mathbb{R}$ are the *Fourier coefficients,*

$$a_k=\frac{2}{T}\int_a^{a+T}x(t)\cos\left(k\frac{2\pi}{T}t\right)dt;\qquad k=0,1,2,\ldots$$

$$b_k=\frac{2}{T}\int_a^{a+T}x(t)\sin\left(k\frac{2\pi}{T}t\right)dt;\qquad k=1,2,\ldots$$

Complex Fourier series for a function $x(t)$ defined on interval $[a,\ a+T]$:

$$x(t)=\sum_{k=-\infty}^{\infty}C_k e^{jk\frac{2\pi}{T}t}$$

where $C_k\in\mathbb{C}$ are the *complex Fourier coefficients:*

$$C_k=\frac{1}{T}\int_a^{a+T}x(t)e^{jk\frac{2\pi}{T}t}\,dt;\qquad k=0,\pm1,\pm2,\ldots$$

Relation between coefficients:

$$C_k=\begin{cases}0.5(a_k-jb_k), & k=1,2,3,\ldots\\ 0.5a_0, & k=0\\ 0.5(a_{-k}+jb_{-k}), & k=-1,-2,-3,\ldots\end{cases}$$

Amplitude spectrum: $\left|C_k\right|=0.5\sqrt{a_k^2+b_k^2}$

Phase spectrum: $\angle C_k=\tan^{-1}\left(\frac{b_k}{a_k}\right)$

Power spectrum: $\left|C_k\right|^2=0.25(a_k^2+b_k^2)$

Parseval's theorem: $\frac{1}{T}\int_a^{a+T}\left|x(t)\right|^2dt=\sum_{k=-\infty}^{\infty}\left|C_k\right|^2$

F.3 Discrete Fourier Transform (DFT)

$$C_k = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}; \qquad k = 0, 1, 2, \ldots, N-1$$

Inverse DFT is defined as

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} C_k e^{j\frac{2\pi}{N}kn}; \qquad n = 0, 1, 2, \ldots, N-1$$

F.4 Convolution Integral

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau$$

F.5 Convolution Sum

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Commutativity of convolution integral and sum:

$$x(t) * h(t) = h(t) * x(t)$$
$$x[n] * h[n] = h[n] * x[n]$$

F.6 Generalized Functions

Dirac Delta

Definition: $\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases}$ and $\int_{-\infty}^{\infty} \delta(t)\, dt = 1.$

1	$\delta(-t) = \delta(t)$
2	$\delta(at) = \frac{\delta(t)}{\lvert a \rvert}$
3	$\delta(t-\tau) = \begin{cases} \infty, & t = \tau \\ 0, & t \neq \tau \end{cases}$
4	$x(t)\delta(t-\tau) = x(\tau)\delta(t-\tau)$

5	$\int_{-\infty}^{\infty} x(t)\delta(t-\tau)\, dt = x(\tau)$
6	$x(t) * \delta(t-\tau) = x(t-\tau)$
7	$x(t) * \dot{\delta}(t) = \dot{x}(t)$
8	$\delta(t) = \frac{d}{dt}\mathcal{U}(t)$

Kronecker Delta

Definition: $\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}.$

1	$\delta[-n] = \delta[n]$
2	$\delta[an] = \delta[n]; \quad a \in \mathbb{Z} \setminus 0$
3	$\delta[n-k] = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$

4	$\sum_{k=-\infty}^{\infty} \delta[n-k]a[k] = a[n]$
5	$\sum_{k=-\infty}^{\infty} \delta[n-k]\,\delta[k-j] = \delta[n-j]$

F.7 Signal Energy and Power

Energy of $x(t)$: $E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\, dt$	Energy of $x[n]$: $E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
Average power of $x(t)$: $P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \lvert x(t)\rvert^2\, dt$	Average power of $x[n]$: $P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} \lvert x[n]\rvert^2$
Average power of T-periodic signal ($\forall a$): $P_T = \frac{1}{T} \int_a^{a+T} \lvert x(t)\rvert^2\, dt$	Average power of N-periodic signal ($\forall a$): $P_N = \frac{1}{N} \sum_{n=a+1}^{a+N} \lvert x[n]\rvert^2$

F.8 Fourier Transform (FT)

FT Definition

Fourier transform $X(\omega) = \mathcal{F}\{x(t)\}$	Inverse Fourier transform $x(t) = \mathcal{F}^{-1}\{X(\omega)\}$
$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt,\ -\infty < \omega < \infty$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$

FT Properties

	Property Name	Signal	Transform
1	Linearity	$ax(t) + by(t)$	$aX(\omega) + bY(\omega)$
2	Time shift	$x(t-a);\quad a \in \mathbb{R}$	$X(\omega) e^{-j\omega a}$
3	Time scaling	$x(at);\quad a \neq 0$	$\frac{1}{\lvert a\rvert} X\left(\frac{\omega}{a}\right)$
4	Time reversal	$x(-t)$	$X(-\omega)$
5	Multiply by t^n	$t^n x(t);\quad n = 1, 2, \ldots$	$j^n \frac{d^n}{d\omega^n} X(\omega)$
6	Multiply by $e^{j\omega_0 t}$	$x(t) e^{j\omega_0 t};\quad \omega_0 \in \mathbb{R}$	$X(\omega - \omega_0)$
7	Multiply by $\cos\omega_0 t$	$x(t)\cos\omega_0 t$	$\frac{1}{2}[X(\omega+\omega_0) + X(\omega-\omega_0)]$
8	Multiply by $\sin\omega_0 t$	$x(t)\sin\omega_0 t$	$\frac{j}{2}[X(\omega+\omega_0) - X(\omega-\omega_0)]$
9	Derivative	$\frac{d^n}{dt^n} x(t)$	$(j\omega)^n X(\omega)$
10	Integral	$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\frac{1}{j\omega} X(\omega) + \pi X(0)\delta(\omega)$
11	Convolution	$x(t) * y(t)$	$X(\omega)Y(\omega)$
12	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} X(\omega) * Y(\omega)$
13	Duality	$X(t)$	$2\pi x(-\omega)$
14	Parseval's theorem	$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lvert X(\omega)\rvert^2\, d\omega$	

FT Common Pairs

1	$\delta(t)$	1
2	$\delta(t-a);\quad a\in\mathbb{R}$	$e^{-j\omega a}$
3	$\mathcal{U}(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$
4	$\mathcal{U}(t)-0.5$	$\frac{1}{j\omega}$
5	$e^{-at}\mathcal{U}(t);\quad a>0$	$\frac{1}{a+j\omega}$
6	1	$2\pi\delta(\omega)$
7	$e^{j\omega_0 t};\quad \omega_0\in\mathbb{R}$	$2\pi\delta(\omega-\omega_0)$
8	$\cos(\omega_0 t)$	$\pi[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]$
9	$\sin(\omega_0 t)$	$j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$
10	$\cos(\omega_0 t+\theta)$	$\pi[e^{-j\theta}\delta(\omega+\omega_0)+e^{j\theta}\delta(\omega-\omega_0)]$
11	$\sin(\omega_0 t+\theta)$	$j\pi[e^{-j\theta}\delta(\omega+\omega_0)-e^{j\theta}\delta(\omega-\omega_0)]$
12	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$
13	$\mathcal{U}(t+a)-\mathcal{U}(t-a)$	$\frac{2\sin(a\omega)}{\omega}$
14	$\frac{\sin(\omega_0 t)}{\pi t}$	$\mathcal{U}(\omega+\omega_0)-U(\omega-\omega_0)$
15	$\text{rect}(t)$	$\text{sinc}_\pi\left(\frac{\omega}{2\pi}\right)$
16	$\text{sinc}_\pi(t)$	$\text{rect}\left(\frac{\omega}{2\pi}\right)$
17	$\text{tri}(t)$	$\left(\text{sinc}_\pi\left(\frac{\omega}{2\pi}\right)\right)^2$
18	$\left(\text{sinc}_\pi\left(\frac{t}{2\pi}\right)\right)^2$	$\text{tri}\left(\frac{\omega}{2\pi}\right)$

F.9 Partial Fractions Expansion

Given a general rational transfer function with real distinct poles, repeated poles, and complex poles:

$$H(s)=\frac{N(s)}{(s+p_1)\cdots(s+p_n)(s+p_m)^r(s^2+as+b)}=\frac{K_1}{s+p_1}+\cdots+\frac{K_n}{s+p_n}+$$
$$+\frac{K_{m1}}{(s+p_m)^r}+\frac{K_{m2}}{(s+p_m)^{r-1}}+\cdots+\frac{K_{mr}}{s+p_m}+$$
$$+\frac{K_l s+K_{l+1}}{s^2+as+b}$$

$$K_i=\lim_{s\to -p_i}(s+p_i)H(s);\quad i=1,\ldots,n$$

$$K_{mi}=\lim_{s\to -p_m}\left(\frac{1}{(i-1)!}\frac{d^{i-1}}{ds^{i-1}}((s+p_m)^r H(s))\right);\quad i=1,\ldots,r$$

F.10 Laplace Transform (LT)

LT Definition

Assume that $\forall(t<0)$: $f(t)=0$.

Laplace transform $F(s)=\mathcal{L}\{f(t)\}$	Inverse Laplace transform $f(t)=\mathcal{L}^{-1}\{F(s)\}$
$F(s)=\int_{0-}^{\infty} f(t)e^{-st}\,dt$	$f(t)=\frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} F(s)e^{st}ds$

LT Properties

	Property Name	Signal	Transform
1	Linearity	$af_1(t)+bf_2(t)$	$aF_1(s)+bF_2(s)$
2	Right shift in time	$f(t-a)\mathcal{U}(t-a);\quad a>0$	$F(s)e^{-as}$
3	Time scaling	$f(at);\quad a>0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
4	Multiply by t^n	$t^n f(t);\quad n=1,2,\ldots$	$(-1)^n\frac{d^n}{ds^n}F(s)$
5	Multiply by e^{at}	$f(t)e^{at};\quad a\in\mathbb{C}$	$F(s-a)$
6	Multiply by $\cos\omega_0 t$	$f(t)\cos\omega_0 t$	$\frac{1}{2}[F(s+j\omega_0)+F(s-j\omega_0)]$
7	Multiply by $\sin\omega_0 t$	$f(t)\sin\omega_0 t$	$\frac{j}{2}[F(s+j\omega_0)-F(s-j\omega_0)]$
8	Derivative	$\dot{f}(t)$	$sF(s)-f(0)$
9	Second derivative	$\ddot{f}(t)$	$s^2F(s)-sf(0)-\dot{f}(0)$
10	nth derivative	$f^{(n)}(t)$	$s^nF(s)-\sum_{k=1}^{n}s^{n-k}f^{(k-1)}(0)$
11	Integral	$\int_0^t f(\tau)\,d\tau$	$\frac{1}{s}F(s)$
12	Convolution	$f_1(t)*f_2(t)$	$F_1(s)F_2(s)$
13	Initial-value theorem	$f(0)=\lim_{s\to\infty}(sF(s))$	
14	Final-value theorem	If $\lim_{t\to\infty}f(t)$ exists, then $\lim_{t\to\infty}f(t)=\lim_{s\to 0}(sF(s))$	

LT Common Pairs

1	$\delta(t)$	1
2	$\delta(t-a);\quad a>0$	e^{-as}
3	$\mathcal{U}(t)$	$\frac{1}{s}$
4	$\mathcal{U}(t)-\mathcal{U}(t-a);\quad a>0$	$\frac{1-e^{-as}}{s}$
5	$e^{-at}\mathcal{U}(t);\quad a\in\mathbb{C}$	$\frac{1}{s+a}$
6	$t^n e^{-at}\mathcal{U}(t);\quad n=1,2,\ldots$	$\frac{n!}{(s+a)^{n+1}}$
7	$(1-at)e^{-at}\mathcal{U}(t)$	$\frac{s}{(s+a)^2}$
8	$\cos(at)\,\mathcal{U}(t)$	$\frac{s}{s^2+a^2}$
9	$\sin(at)\,\mathcal{U}(t)$	$\frac{a}{s^2+a^2}$
10	$\frac{-e^{-at}+e^{-bt}}{a-b}\mathcal{U}(t)$	$\frac{1}{(s+a)(s+b)}$
11	$\frac{ae^{-at}-be^{-bt}}{a-b}\mathcal{U}(t)$	$\frac{s}{(s+a)(s+b)}$
12	$\frac{(c-b)e^{-at}+(a-c)e^{-bt}+(b-a)e^{-ct}}{(a-b)(b-c)(c-a)}\mathcal{U}(t)$	$\frac{1}{(s+a)(s+b)(s+c)}$
13	$\frac{-a(c-b)e^{-at}-b(a-c)e^{-bt}-c(b-a)e^{-ct}}{(a-b)(b-c)(c-a)}\mathcal{U}(t)$	$\frac{s}{(s+a)(s+b)(s+c)}$
14	$\frac{-a(c-b)e^{-at}-b(a-c)e^{-bt}-c(b-a)e^{-ct}}{(a-b)(b-c)(c-a)}\mathcal{U}(t)$	$\frac{s^2}{(s+a)(s+b)(s+c)}$
15	$\frac{e^{-at}-e^{-bt}+(a-b)te^{-bt}}{(a-b)^2}\mathcal{U}(t)$	$\frac{1}{(s+a)(s+b)^2}$
16	$\frac{-ae^{-at}+ae^{-bt}-b(a-b)te^{-bt}}{(a-b)^2}\mathcal{U}(t)$	$\frac{s}{(s+a)(s+b)^2}$
17	$\frac{a^2e^{-at}-b(2a-b)e^{-bt}+b^2(a-b)te^{-bt}}{(a-b)^2}\mathcal{U}(t)$	$\frac{s^2}{(s+a)(s+b)^2}$
18	$e^{-bt}\cos(at)\,\mathcal{U}(t)$	$\frac{s+b}{(s+b)^2+a^2}$
19	$e^{-bt}\sin(at)\,\mathcal{U}(t)$	$\frac{a}{(s+b)^2+a^2}$

F.11 Ordinary Differential Equations

Let $y^{(n)}+a_{n-1}y^{(n-1)}+\cdots+a_1y^{(1)}+a_0y=f(t)$ be the ordinary differential equation (ODE), where $f(t)$ is some function of time.

We apply the Laplace transform to both sides of the equation $\dot{y}(t)\xrightarrow{\text{Laplace}}sY(s)-y(0)$ and get an algebraic equation. We then solve this algebraic equation with regard to s to find $Y(s)$. Finally, we apply the inverse Laplace transform to get the solution $y(t)$. Similarly, we can solve discrete-time difference equations using the Z-transform.

A general solution of a linear homogeneous equation [where $f(t) = 0$] is $y(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} + \cdots + C_n e^{s_n t}$, where $C_1, C_2, \ldots, C_n$ are constants, and $s_1, s_2, \ldots, s_n$ are the solutions of the polynomial equation $s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$. The constants C_i are found from the initial conditions of the ODE.

The general solution of a first-order ODE: $\frac{\partial x(t)}{\partial t} + p(t)x(t) = q(t); x(t_0) = x_0$ is

$$x(t) = \exp\left(-\int_{t_0}^{t} p(\tau)d\tau\right)\left(x_0 + \int_{t_0}^{t}\left[q(\tau)\exp\left(\int_{t_0}^{\tau} p(\sigma)d\sigma\right)\right]d\tau\right)$$

F.12 Z-Transform

Z-Transform Definition

Assume that $\forall(n < 0)$: $x[n] = 0$.

Laplace transform $X(z) = Z\{x(t)\}$	Inverse Laplace transform $x[n] = Z^{-1}\{X(z)\}$
$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$	$x[n] = \frac{1}{2\pi j}\int_{\substack{\text{convergence}\\ \text{circle}}} X(z)z^{k-1}dz$

Z-Transform Properties

	Property Name	Signal	Transform
1	Linearity	$ax[n] + by[n]$	$aX(z) + bY(z)$
2	Right shift in time	$x[n-a]; \quad a \in \mathbb{N}$	$z^{-a}X(z)$
3	Left shift in time	$x[n+a]; \quad a \in \mathbb{N}$	$z^a X(z) - \sum_{k=0}^{a-1} x[k]z^{a-k}$
4	Multiply by n	$nx[n]$	$-z\frac{d}{dz}X(z)$
5	Multiply by n^2	$n^2x[n]$	$z\frac{d}{dz}X(z) + z^2\frac{d^2}{dz^2}X(z)$
6	Multiply by a^n	$a^n x[n]$	$X\left(\frac{z}{a}\right)$
7	Multiply by $\cos\omega_0 n$	$x[n]\cos\omega_0 n$	$\frac{1}{2}[X(e^{j\omega_0}z) + X(e^{j\omega_0}z)]$
8	Multiply by $\sin\omega_0 n$	$x[n]\sin\omega_0 n$	$\frac{j}{2}[X(e^{j\omega_0}z) - X(e^{j\omega_0}z)]$
9	Summation	$\sum_{i=0}^{n} x[i]$	$\frac{z}{z-1}X(z)$
10	Convolution	$x[n] * y[n]$	$X(z)Y(z)$
11	Initial-value theorem	$x[0] = \lim_{z\to\infty}(X(z))$	
12	Final-value theorem	If $X(z)$ is rational and the poles of $(z-1)X(z)$ have magnitudes <1, then $\lim_{n\to\infty} x[n] = [(z-1)X(z)]_{z=1}$	

Z-Transform Common Pairs

1	$\delta[n]$	1
2	$\delta[n-a];\quad a=1,2,\ldots$	z^{-a}
3	$\mathcal{U}[n]$	$\frac{z}{z-1}$
4	$\mathcal{U}[n]-\mathcal{U}[n-a];\quad a=1,2,\ldots$	$\frac{z^a-1}{z^{a-1}(z-1)}$
5	$a^{n-k}\,\mathcal{U}[n-k];\quad a\in\mathbb{C},\,k=0,1,2,\ldots$	$\frac{z^{1-k}}{z-a}$
6	$na^n\,\mathcal{U}[n]$	$\frac{az}{(z-a)^2}$
7	$(n+1)\mathcal{U}[n]$	$\frac{z^2}{(z-1)^2}$
8	$n^2a^n\mathcal{U}[n]$	$\frac{az(z+a)}{(z-a)^3}$
9	$\frac{a^{n+k}-b^{n+k}}{a-b}\mathcal{U}[n+k-1];\quad k\in\mathbb{Z}$	$\frac{z^{k+1}}{(z-a)(z-b)}$
10	$\left(\frac{(a-b)(n+k)a^{n+k-1}-a^{n+k}+b^{n+k}}{(a-b)^2}\right);\quad k=2,1,0,\ldots$	$\frac{z^{k+1}}{(z-a)^2(z-b)}$
11	$\cos[bn]\,\mathcal{U}[n]$	$\frac{z(z-\cos b)}{z^2-(2\cos b)z+1}$
12	$\sin[bn]\,\mathcal{U}[n]$	$\frac{z\sin b}{z^2-(2\cos b)z+1}$
13	$a^n\cos[bn]\,\mathcal{U}[n]$	$\frac{z(z-a\cos b)}{z^2-(2a\cos b)z+a^2}$
14	$a^n\sin[bn]\,\mathcal{U}[n]$	$\frac{za\sin b}{z^2-(2a\cos b)z+a^2}$

G Linear Algebra

A rectangular table of $m\times n$ numbers is called a matrix:

$$A_{m\times n}=(a_{ij})=\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where m is the number of rows and n is the number of columns. The numbers a_{ij} ($i=1,2,\ldots,m$; $j=1,2,\ldots,n$) are called matrix elements. The first index i points to the row number and the second index j points to the column number where this element is located.

If the number of rows equals the number of columns ($m=n$), the matrix is called square of the order n.

G.1 Special Matrices

A diagonal matrix has zeros everywhere outside of the main diagonal:

$$A = \text{diag}(a_{11}, a_{22}, \ldots, a_{nn}) = \begin{pmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{pmatrix} = (a_{ij}\delta_{ij})$$

when δ_{ij} is defined by: $\delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{else} \end{cases}$

An *identity matrix* is a special kind of a diagonal matrix with 1s on the main diagonal ($a_{ii} = 1$) and denoted by I.

An *upper (lower) triangular matrix* is the matrix with zero elements below (above) the main diagonal.

G.2 Matrix Addition and Subtraction

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{pmatrix} \pm \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & \cdots & a_{1n} \pm b_{1n} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & \cdots & a_{2n} \pm b_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} \pm b_{m1} & a_{m2} \pm b_{m2} & \cdots & a_{mn} \pm b_{mn} \end{pmatrix}$$

G.3 Matrix Determinant

$$\det(A_{2\times 2}) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\det(A_{3\times 3}) = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$\det(A) = |A| = a_{11} \begin{vmatrix} a_{22} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{n2} & \cdots & a_{nn} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} & \cdots & a_{2n} \\ a_{31} & a_{33} & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n3} & a_{n4} & \cdots & a_{nn} \end{vmatrix} +$$

$$+ a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n4} & \cdots & a_{nn} \end{vmatrix} - \cdots$$

You can reduce the determinant's order from n to $n-1$ using the approach above to simplify the computation.

Determinant properties:

1. Switching two consecutive rows (columns) changes the sign of the determinant.
2. If there are any linearly dependent rows (columns) then the determinant is zero.
3. Adding a linear combination of rows (columns) to some row (column) will not change the determinant.
4. $\det(kA) = k^n \det(A)$, when k is a scalar.
5. For square matrices A and B: $\det(AB) = \det(A)\det(B)$.
6. $\det(I_{m\times m} + A_{m\times n}B_{n\times m}) = \det(I_{n\times n} + B_{n\times m}A_{m\times n})$.
7. $\det(A^{-1}) = \det(A)^{-1}$.

G.4 Singular Matrix

If $\det(A) = 0$, A is called singular (or noninvertible). The matrix is singular if and only if it has linearly dependent rows (or columns).

G.5 Regular Matrix

Nonsingular matrices are called regular matrices.

NOTE The terms "determinant," "regular," and "singular" are defined only for square matrices.

G.6 Transposed Matrix

$A^T = (a_{ji})$ is the transposed matrix $A = (a_{ij})$. Note that in the transposed matrix the rows are switched with the columns.

Properties:

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(kA)^T = kA^T$, k is scalar
4. $(AB)^T = B^T A^T$
5. $\det(A^T) = \det(A)$
6. $\det(sI - A^T) = \det(sI - A)$ (eigenvalues of A and A^T are the same)

G.7 Symmetric Matrix

$$A^T = A$$

G.8 Antisymmetric Matrix

$$A^T = -A$$

G.9 Matrix Multiplication

$$A_{m\times n}\cdot B_{n\times p}=C_{m\times p}=(c_{ij})=\left(\sum_{k=1}^{n}a_{ik}b_{kj}\right);\qquad (i=1,2,\ldots,m;\ j=1,2,\ldots,p)$$

Caution! The matrix multiplication is not always commutative, that is, it could be that $AB\neq BA$. For example, $\begin{pmatrix}3&0\\7&0\end{pmatrix}\begin{pmatrix}1&2\\1&2\end{pmatrix}=\begin{pmatrix}3&6\\7&14\end{pmatrix}\neq\begin{pmatrix}17&0\\17&0\end{pmatrix}=\begin{pmatrix}1&2\\1&2\end{pmatrix}\begin{pmatrix}3&0\\7&0\end{pmatrix}$.

Caution! If $AC=BC$, it does not mean that $A=B$. For example, $\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix}=\begin{pmatrix}1&2\\4&3\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix}$, but $\begin{pmatrix}1&2\\3&4\end{pmatrix}\neq\begin{pmatrix}1&2\\4&3\end{pmatrix}$.

G.10 Powers of the Square Matrix

$$A^k=\underbrace{AA\ldots A}_{k\text{ times}}$$

G.11 Square Root of the Square Matrix (Cholesky Decomposition)

Algorithm for finding the matrix L so that $LL^T=A$ for a positive symmetric matrix A (see definitions below):

$$L_{ii}=\sqrt{A_{ii}-\sum_{k=1}^{i-1}L_{ik}^2}\quad i=1,\ldots,n\qquad\qquad L_{ji}=\frac{1}{L_{ii}}\left(A_{ij}-\sum_{k=1}^{i-1}L_{ik}L_{jk}\right)\quad j=i+1,\ldots,n$$

G.12 Matrix Exponent

For a scalar power, the exponent could be developed into the following Taylor series:

$$e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots$$

For $x=at$ (a is scalar):

$$e^{at}=1+at+\frac{(at)^2}{2!}+\frac{(at)^3}{3!}+\frac{(at)^4}{4!}+\cdots$$

Similarly, we could define for a matrix A:

$$e^{At}=I+At+\frac{(At)^2}{2!}+\frac{(At)^3}{3!}+\frac{(At)^4}{4!}+\cdots$$

Note The exponent of $n\times n$ matrix A is $n\times n$ matrix too.

If we take the Laplace transform from both sides for a large enough $|s|$:

$$\mathcal{L}\{e^{At}\}=\frac{I}{s}+\frac{A}{s^2}+\frac{A^2}{s^3}+\frac{A^3}{s^4}+\frac{A^4}{s^5}+\cdots=\frac{1}{s}\left(I+\left(\frac{A}{s}\right)+\left(\frac{A}{s}\right)^2+\left(\frac{A}{s}\right)^3+\left(\frac{A}{s}\right)^4+\cdots\right)$$

On the other hand, for scalar x, we know that the Taylor series of $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ and for some matrix C it would be $(I-C)^{-1} = I + C + C^2 + C^3 + \cdots$ if this series is converging. Thus, for $C = -\frac{A}{s}$, we get $\mathcal{L}\{e^{At}\} = \frac{1}{s}\left(I - \frac{A}{s}\right)^{-1} = (sI - A)^{-1}$. Now, if we take the inverse Laplace transform from both sides, we get

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

NOTE The solution of $\dot{x}(t) = Ax(t)$ matrix differential equation is the vector $x(t) = e^{At}x(0)$.

Properties:

- $p(A)e^{At} = e^{At}p(A)$, for any polynomial p
- $e^{A^T t} = (e^{At})^T$
- $e^{-At} = (e^{At})^{-1}$
- $\frac{d}{dt}e^{At} = Ae^{At}$

CAUTION! It is not always true that $e^A e^B = e^{A+B}$ (true only when $AB = BA$). Here is an example: $e^{\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)}e^{\left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)} = \begin{pmatrix}0.54 & 1.38\\ -0.84 & -0.3\end{pmatrix}$ and $e^{\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)+\left(\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right)} = \begin{pmatrix}0.16 & 1.4\\ -0.7 & 0.16\end{pmatrix}$.

G.13 Matrix Rank

rank(A) is the number of linearly independent rows (columns).

Rows $r_1, r_2, \ldots, r_n$ are linearly independent if their linear combination $\alpha_1 r_1 + \alpha_2 r_2 + \cdots + \alpha_n r_n$ is not equal to zero for any α_i; $i = 1, \ldots, n$ (except the trivial combination $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$).

Properties:

- rank(AB) ≤ min(rank(A), rank(B)).
- Square matrix A has a full rank if and only if A is regular.

G.14 Inverse of a Matrix

The inverse of a matrix A is denoted by A^{-1} and satisfies

$$A(A^{-1}) = (A^{-1})A = I$$

Properties:

1. $(A^{-1})^{-1} = A$
2. $(A^{-1})^T = (A^T)^{-1}$
3. $(AB)^{-1} = B^{-1}A^{-1}$
4. Lemma of matrix inverse: If A and C are regular and $A + BCD$ exists and is invertible, then: $(A + BCD)^{-1} = A^{-1} - A^{-1}B(DA^{-1}B + C^{-1})^{-1}DA^{-1}$
5. Given $A_{n\times n}$, $B_{n\times m}$, $C_{m\times n}$: $(I - C(sI - A)^{-1}B)^{-1} = I + C(sI - A - BC)^{-1}B$

Calculate inverse matrix:

2×2:

$$A=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1}=\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

3×3:

$$A=\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \Rightarrow A^{-1}=\frac{1}{\det(A)}\begin{pmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} & -\begin{vmatrix} b & c \\ h & i \end{vmatrix} & \begin{vmatrix} b & c \\ e & f \end{vmatrix} \\ -\begin{vmatrix} d & f \\ g & i \end{vmatrix} & \begin{vmatrix} a & c \\ g & i \end{vmatrix} & -\begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ \begin{vmatrix} d & e \\ g & h \end{vmatrix} & -\begin{vmatrix} a & b \\ g & h \end{vmatrix} & \begin{vmatrix} a & b \\ d & e \end{vmatrix} \end{pmatrix}$$

CAUTION! In solving matrix equations, it is prohibited to reduce matrices on both sides. To simplify the equation, you need to multiply both sides of the equation by some inverse matrix from the left or from the right, and then use $AA^{-1}=I$.

G.15 Trace of a Matrix

For a square matrix $A_{n\times n}=(a_{ij})$, we define: $tr(A)=\sum_{i=1}^{n}a_{ii}$, the trace of the matrix A.

Properties:

- $tr(A+B)=tr(A)+tr(B)$
- $tr(AB)=tr(BA)$

Example

To compute the trace of the following matrix: $\begin{pmatrix} 1 \\ 2 \\ 0 \\ 4 \\ 0 \end{pmatrix}\cdot(1 \quad 2 \quad 3 \quad 0 \quad 5)$, we could compute the 5×5 matrix first and then add all elements on the main diagonal. Instead, it is much more convenient to use the property $tr(AB)=tr(BA)$ and get an element-wise multiplication of those two vectors: $tr\left(\begin{pmatrix} 1 \\ 2 \\ 0 \\ 4 \\ 0 \end{pmatrix}(1 \quad 2 \quad 3 \quad 0 \quad 5)\right)=tr\left((1 \quad 2 \quad 3 \quad 0 \quad 5)\begin{pmatrix} 1 \\ 2 \\ 0 \\ 4 \\ 0 \end{pmatrix}\right)=$

$1\cdot 1+2\cdot 2+0\cdot 3+4\cdot 0+0\cdot 5=5$.

G.16 Orthogonal Matrix

$A^T = A^{-1}$ (A is a real matrix).

G.17 Orthonormal Matrix

The orthonormal matrix is an orthogonal matrix with the columns normalized to the norm 1.

G.18 Eigenvalues and Eigenvectors of a Square Matrix

For a matrix $A_{n\times n}$, we define the *eigenvalues* as the scalar solutions s_i of the equation $Ax_i = s_i x_i$ for some vector x_i. The vector x_i that belongs to the eigenvalue s_i is called *eigenvector*.

Computation: The polynomial $\det(sI - A)$ is called the *characteristic polynomial* $\Delta(s)$ of A, and its roots are the eigenvalues of A.

By solving the equation $\det(sI - A) = 0$ we find all the eigenvalues $\{s_i\}$, and then by solving the system of equations $Ax_i = s_i x_i$ we find all the eigenvectors $\{x_i\}$.

Diagonalization matrix: Construct a diagonalization transform matrix T from the eigenvectors as its columns. You may apply this similarity transform to the matrix A to get: $T^{-1}AT = \text{diag}\{s_i\}$.

Properties:

1. $tr(A) = \sum_{i=1}^{n} s_i$.
2. $\det(A) = \prod_{i=1}^{n} s_i$.
3. $A - s_i I$ is singular.
4. For 2×2 matrices: $\Delta(s) = \det(sI - A) = s^2 - (tr(A))s + \det(A)$.
5. Any real symmetric matrix A is diagonizable, all its eigenvalues are real, and all its eigenvectors are orthogonal.
6. $s = 0$ is an eigenvalue of A if and only if A is singular (noninvertible).
7. $s \neq 0$ is an eigenvalue of A if and only if subtracting s from the main diagonal will reduce the $\text{rank}(A)$.
8. All eigenvalues of upper or lower triangular matrices are located on the main diagonal.
9. If $f(A)$ is an analytic function of A that can be written as polynomial series (most functions are …), then for any invertible transform T, $T^{-1}f(A)T = f(T^{-1}AT)$. This is an effective technique to compute complicated matrix functions.
10. A and A^T have the same eigenvalues.

Example

Let's say we want to compute $\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}^{2000}$ without using a computer. To solve that problem, we will define $A = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$ and compute its eigenvalues s_i and eigenvectors x_i.

This will allow us to find a diagonalizing similarity transform T such that $T^{-1}AT$ is diagonal. Then, we will use the identity $A^{2000} = (T\,\mathrm{diag}(s_i)T^{-1})^{2000}$ to compute the matrix.

To compute eigenvalues, $\det(sI - A) = \begin{vmatrix} s-1 & -2 \\ 0 & s-2 \end{vmatrix} = (s-1)(s-2) = 0$, thus $s_1 = 1$ and $s_2 = 2$. We could get those values using property 8 of eigenvalues without any computations. Now, we need to compute the eigenvectors. For that, we solve the system of equations $Ax_i = s_i x_i$ for each s_i:

$$\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = s_1\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \begin{cases} x_1 + 2x_2 = x_1 \\ 2x_2 = x_2 \end{cases} \Rightarrow x_2 = 0$$

Note that we should always get an infinite number of solutions if the matrix is diagonalizable. In that case, we can choose any x_1, for example, $x_1 = 1$, and get $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Similarly,

$$\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = s_2\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \begin{cases} x_1 + 2x_2 = 2x_1 \\ 2x_2 = 2x_2 \end{cases} \Rightarrow x_1 = 2x_2$$

Again, we have one degree of freedom and we can choose any x_2. For example, $x_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

The diagonalizing transform is $T = (x_1 | x_2) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, thus $T^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$, and the diagonal matrix of eigenvalues is $D = \mathrm{diag}(s_i) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

Finally, $A^{2000} = (TDT^{-1})^{2000} = \underbrace{(TDT^{-1})(TDT^{-1})(TDT^{-1})\cdots(TDT^{-1})}_{2000 \text{ times}} = TD^{2000}T^{-1} =$

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^{2000}\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1^{2000} & 0 \\ 0 & 2^{2000} \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2^{2001}-2 \\ 0 & 2^{2000} \end{pmatrix}.$$

Cayley-Hamilton Theorem

If $\det(sI - A) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n$, then $A^n + \alpha_1 A^{n-1} + \cdots + \alpha_n I = 0$. In other words, matrix A is a zero of its own characteristic polynomial.

The polynomial of the lowest degree $p(s)$ such that $p(A) = 0$ is called a *minimal polynomial* of A.

G.19 Similar Matrices

Matrices A and B are *similar*, if there exists an invertible transform T such that $A = T^{-1}BT$ (alternatively, $TA = BT$).

Properties:

- Similar matrices have the same eigenvalues.
- Matrix A is diagonizable if and only if there exists the invertible transformation T such that $T^{-1}AT$ is diagonal, and the columns of T are the eigenvectors of A.
- A is similar to A^T.

Example

We will find a similarity transform between A and A^T if they are diagonalizable. Let's assume that T_1 is a diagonalization transform of A and T_2 is a diagonalization transform of A^T. Since both A and A^T have the same eigenvalues diagonal matrix D, $T_1^{-1}AT_1 = D = T_2^{-1}A^TT_2$. Now, we multiply both sides of the equation by T_1 on the left and T_1^{-1} on the right: $\cancel{T_1}\cancel{T_1^{-1}}A\cancel{T_1}\cancel{T_1^{-1}} = A = T_1T_2^{-1}A^TT_2T_1^{-1}$. So, if we define the similarity transform $P = T_2T_1^{-1}$, then $A = P^{-1}A^TP$.

G.20 Positive Definite and Semidefinite Matrices

A real symmetric matrix $A_{n\times n}$ is called *positive definite* if for any vector $x \neq 0$ the quadratic form is positive, that is, $x^TAx = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j > 0$. We write that as $A \succ 0$.

A real symmetric matrix $A_{n\times n}$ is called *positive semidefinite* if for any vector $x \neq 0$ the quadratic form is positive, that is, $x^TAx = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j \geq 0$.

Properties:

1. *Sylvester's Criterion:* If for the symmetric matrix A, all the main minors are positive $a_{11} > 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{vmatrix} > 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{vmatrix} > 0 \ldots$, then A is positive definite.
2. If all the main minors are nonnegative, then symmetric A is positive semidefinite.
3. A square matrix is symmetric and positive definite if and only if there exists invertible matrix D such that $A = D^TD$.
4. A square matrix of the order n is symmetric and positive semidefinite if and only if there exists matrix D $(\text{rank}(D) < n)$ such that $A = D^TD$.
5. Matrix A is positive definite (semidefinite) if and only if all its eigenvalues are positive (nonnegative).

G.21 Block Matrices

1. $$\begin{pmatrix} \boxed{A_{11}} & \boxed{A_{12}} \\ \boxed{A_{21}} & \boxed{A_{22}} \end{pmatrix}\begin{pmatrix} \boxed{B_{11}} & \boxed{B_{12}} \\ \boxed{B_{21}} & \boxed{B_{22}} \end{pmatrix} = \begin{pmatrix} \boxed{A_{11}B_{11} + A_{12}B_{21}} & \boxed{A_{11}B_{12} + A_{12}B_{22}} \\ \boxed{A_{21}B_{11} + A_{22}B_{21}} & \boxed{A_{21}B_{12} + A_{22}B_{22}} \end{pmatrix}$$
2. For square matrices A and D: $\det\begin{pmatrix} \boxed{A} & \boxed{0} \\ \boxed{C} & \boxed{D} \end{pmatrix} = \det\begin{pmatrix} \boxed{A} & \boxed{B} \\ \boxed{0} & \boxed{D} \end{pmatrix} = \det(A)\det(D)$.

Note The eigenvalues of a block-triangular matrix are the eigenvalues of the main diagonal blocks.

3. If all matrices in the blocks exist,

$$\begin{pmatrix} \boxed{A} & \boxed{B} \\ \boxed{C} & \boxed{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boxed{A^{-1} - A^{-1}B(CA^{-1}B - D)^{-1}CA^{-1}} & \boxed{A^{-1}B(CA^{-1}B - D)^{-1}} \\ \boxed{(CA^{-1}B - D)^{-1}CA^{-1}} & \boxed{-(CA^{-1}B - D)^{-1}} \end{pmatrix} =$$

$$= \begin{pmatrix} \boxed{-(BD^{-1}C - A)^{-1}} & \boxed{(BD^{-1}C - A)^{-1}BD^{-1}} \\ \boxed{D^{-1}C(BD^{-1}C - A)^{-1}} & \boxed{D^{-1} - D^{-1}C(BD^{-1}C - A)^{-1}BD^{-1}} \end{pmatrix}$$

4. If A is invertible, then $\det\begin{pmatrix} \boxed{A} & \boxed{B} \\ \boxed{C} & \boxed{D} \end{pmatrix} = \det(A)\det(D - CA^{-1}B)$.

5. If A and D are invertible, then

$$\begin{pmatrix} \boxed{A} & \boxed{0} \\ \boxed{C} & \boxed{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boxed{A^{-1}} & \boxed{0} \\ \boxed{-D^{-1}CA^{-1}} & \boxed{D^{-1}} \end{pmatrix}; \quad \begin{pmatrix} \boxed{A} & \boxed{B} \\ \boxed{0} & \boxed{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boxed{A^{-1}} & \boxed{-A^{-1}BD^{-1}} \\ \boxed{0} & \boxed{D^{-1}} \end{pmatrix}$$

Example

For the following matrix $M = \begin{pmatrix} 9 & -1 & 5 \\ 8 & 3 & 2 \\ 0 & 0 & 3 \end{pmatrix}$, we need to compute $\det(M)$, M^{-1}, and prove that $M \succ 0$.

We divide the matrix into four blocks as follows:

$$M = \left(\begin{array}{cc|c} 9 & -1 & 5 \\ 8 & 3 & 2 \\ \hline 0 & 0 & 3 \end{array}\right)$$

The blocks are $A = \begin{pmatrix} 9 & -1 \\ 8 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$, $C = (0 \quad 0)$, $D = (3)$. Since C is a block of zeros, the matrix is block triangular. Therefore,

$$\det(M) = \det(A)\det(D) = (9 \cdot 3 + 8) \cdot 3 = 105$$

$$M^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{pmatrix} = \left(\begin{array}{cc|c} \frac{1}{35}\begin{pmatrix} 3 & 1 \\ -8 & 9 \end{pmatrix} & & -A^{-1}BD^{-1} \\ \hline 0 & & 1/3 \end{array}\right) = \begin{pmatrix} 3/35 & 1/35 & -0.162 \\ -8/35 & 9/35 & 0.21 \\ 0 & 0 & 1/3 \end{pmatrix}$$

It would be easy to use the Sylvester criterion to check if the matrix is positive definite, but we cannot do it because the matrix is not symmetrical. We need to find the eigenvalues (which are the eigenvalues of the main diagonal blocks for block triangular matrix). These eigenvalues are 3, 5, and 7. All eigenvalues are positive, thus $M \succ 0$.

H Random Variables

Probability is a number between 0 and 1 that shows the odds of some event, and where numbers closer to 1 mean stronger likelihood. In simple cases, probability could be approximated by estimating the ratio between the number of desired outcomes out of all possible outcomes.

H.1 Expected Value (Mean)

Given a random signal X that could have possible values (outcomes) $x_1, x_2, \ldots, x_n, \ldots$ and given probability of each one of those values $p_1, p_2, \ldots, p_n, \ldots$ described by a given distribution, the *expected value* (or mean, or average) for discrete random variables is given by

$$E[X] = \sum_i p_i x_i = p_1 x_1 + p_2 x_2 + \cdots + p_n x_n + \cdots$$

which is the weighted sum of x_i values, where $\sum_i p_i = 1$.

For example, if there are 10 possible values that X can take, and all of them are equally likely, then the expected value will be $E[X] = \sum_i \frac{1}{10} x_i = \frac{1}{10} \sum_i x_i$, which is an arithmetic mean.

For continuous random variables, and a given density function $p(x)$,

$$E[X] = \int_{-\infty}^{\infty} x p(x) dx$$

Properties:

- If $X = Y$, then $E[X] = E[Y]$
- $E[\alpha X + \beta Y] = \alpha E[X] + \beta E[Y]$
- $E\left[\sum_{i=1}^{\infty} X_i\right] = \sum_{i=1}^{\infty} E[X_i]$

H.2 Variance

A *variance* is the amount of deviation from the mean μ. It is defined by

$$\mathrm{Var}(X) = E[(X - \mu)^2]$$

Because of the linearity property of the expected value, it could be shown that

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

A *standard deviation* is a square root of variance:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

H.3 Gaussian Distribution

There are multiple possible distributions available to describe random processes and noises. The most popular is *Gaussian (normal) distribution*, which is given by (in 1D):

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is distribution's mean and σ is distribution's standard deviation. It is denoted by $x \sim N(\mu, \sigma)$.

Multivariate distribution for a vector x is given by

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

where $x = (x_1, x_2, \ldots, x_n)^T$, $\mu = (E[x_1], E[x_2], \ldots, E[x_n])^T$, $\exp(y) = e^y$, Σ is the covariance matrix, and $|\Sigma| = \det(\Sigma)$.

The covariance matrix elements are defined by $(\Sigma_{i,j}) = E[(x_i - E[x_i])(x_j - E[x_j])]$:

$$\Sigma = \mathrm{Cov(x)}$$

$$= \begin{pmatrix} E[(x_1 - E[x_1])^2] & E[(x_1 - E[x_1])(x_2 - E[x_2])] & \cdots & E[(x_1 - E[x_1])(x_n - E[x_n])] \\ E[(x_2 - E[x_2])(x_1 - E[x_1])] & E[(x_2 - E[x_2])^2] & \cdots & E[(x_2 - E[x_2])(x_n - E[x_n])] \\ \vdots & \vdots & \ddots & \vdots \\ E[(x_n - E[x_n])(x_1 - E[x_1])] & E[(x_n - E[x_n])(x_2 - E[x_2])] & \cdots & E[(x_n - E[x_n])^2] \end{pmatrix}$$